LECTURE · HMM IN FINANCE

Hidden Markov Models in Finance

Overview, Stock Prediction and Information-Based Trading

HMM basics → model selection → return regimes → rates carry → PIN / VPIN

Every figure in this deck comes from the original lecture slides.
Worked with hmmlearn on AAPL, KOSPI and EUR/USD futures data.

Markets switch between regimes we never observe directly

Three ideas carry this lecture

1

Prices and flows are emissions of hidden states

A Gaussian HMM lets each hidden state have its own return or order-flow distribution.

2

Information criteria fix the number of states

AIC and BIC trade likelihood against the $N^2 + 2N - 1$ parameters of the model.

3

Decoded states become trading signals

Return regimes, carry risk premia and informed-trader activity all come from the same machinery.

One model, three markets: equities, rates and FX order flow.

01

PART 1 · HMM OVERVIEW

Hidden Markov Model Overview

Markov property, emissions and the three problems

A Markov model depends on the past only through the latest state

Basic definition: transition probability

$$P\Big(Z(t+1) \mid Z(1), \ldots, Z(t)\Big) = P\Big(Z(t+1) \mid Z(t)\Big)$$

Why this matters

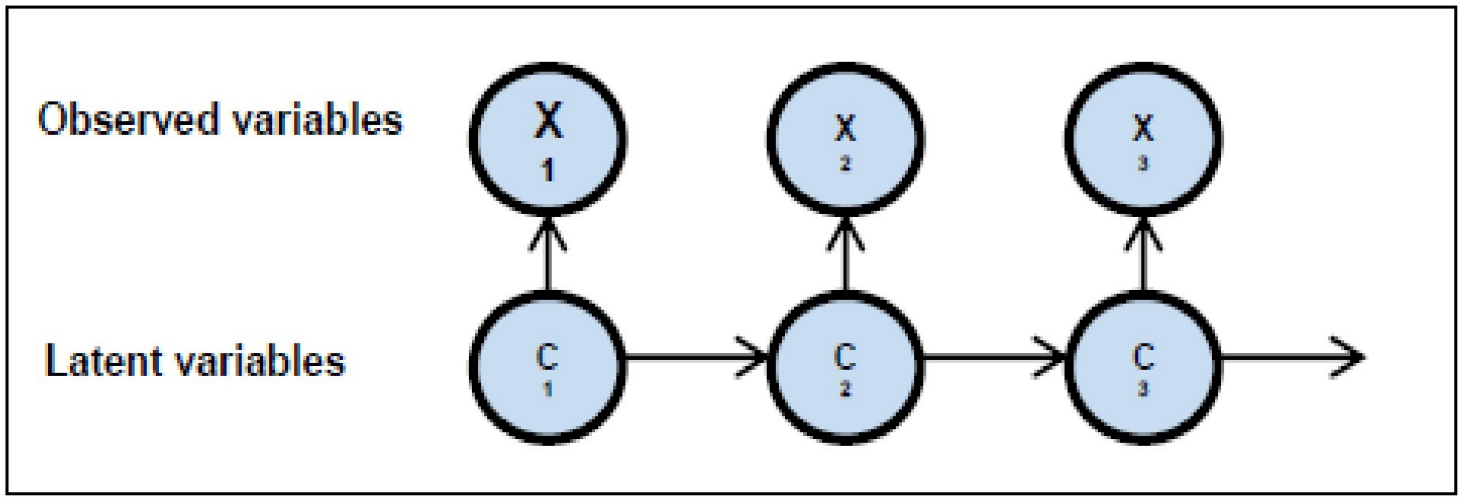
- The whole history Z(1)…Z(t) is summarised by Z(t)
- The model needs only a transition matrix between states

An HMM adds observations that depend only on the current hidden state

Latent variables C_t form a Markov chain; observed X_t hang from them

Two assumptions

- Hidden chain is Markov
- X(t) depends on Z(t) only
- Example on the right



$$P\Big(Z(t) \mid Z(1), \ldots, Z(t-1)\Big) = P\Big(Z(t) \mid Z(t-1)\Big)$$

$$P\Big(X(t) \mid X(1), \ldots, X(t-1), Z(1), \ldots, Z(t)\Big) = P\Big(X(t) \mid Z(t)\Big)$$

Each HMM problem has its own algorithm

Initial state, transition and emission probabilities define the model

Evaluation

Forward-Backward

Given the initial, transition and emission probabilities, compute the probability of the observed sequence X.

Decoding

Viterbi

Find the most likely sequence of hidden states Z.

Learning

Baum-Welch

Given only X, estimate the initial, transition and emission probabilities and the state sequence Z.

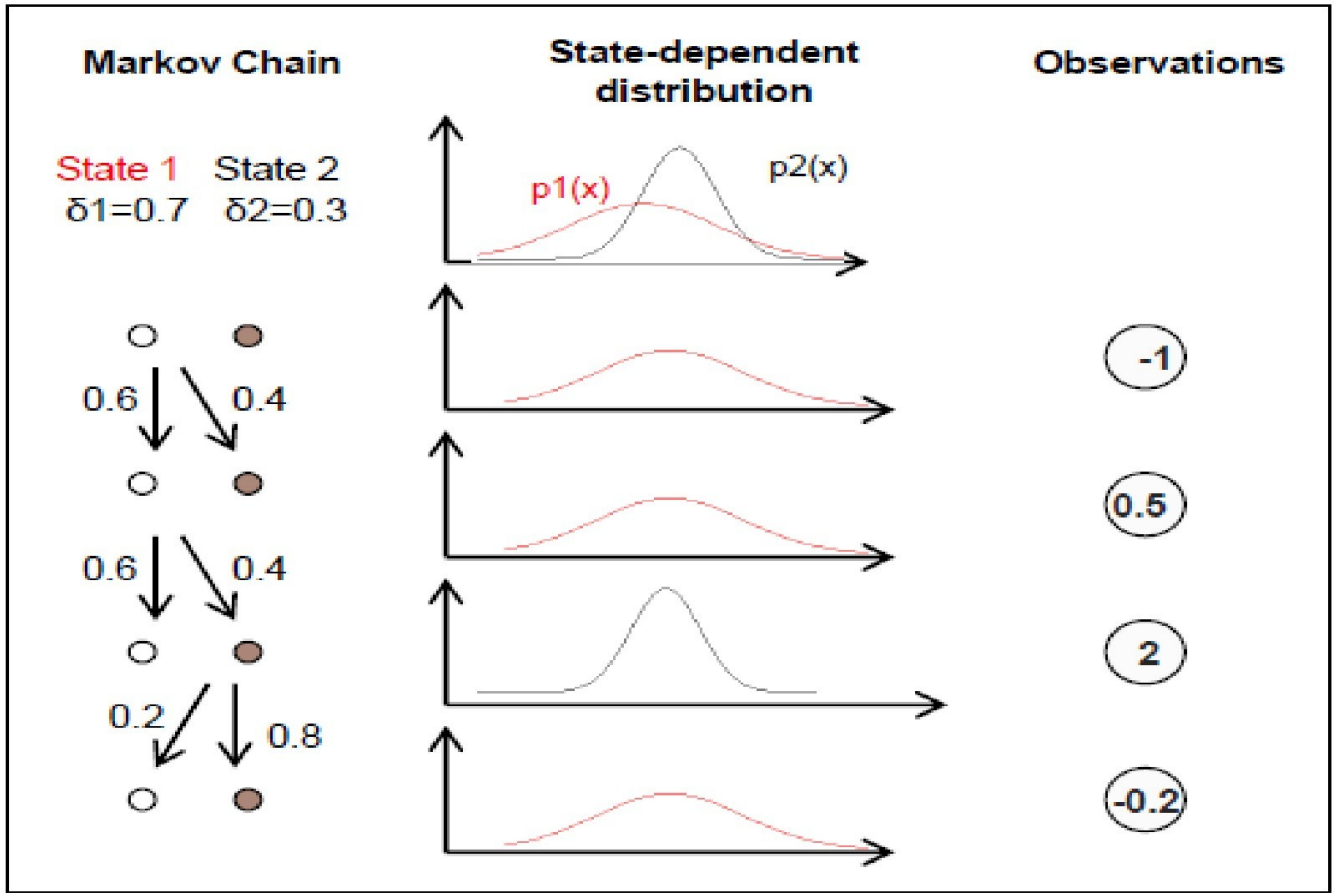
This lecture uses all three through the hmmlearn package.

Each hidden state draws observations from its own distribution

Two states (δ1 = 0.7, δ2 = 0.3) emit returns such as −1, 0.5, 2, −0.2

Read left to right

- Markov chain moves states
- Each state has p(x)
- Observation drawn per step

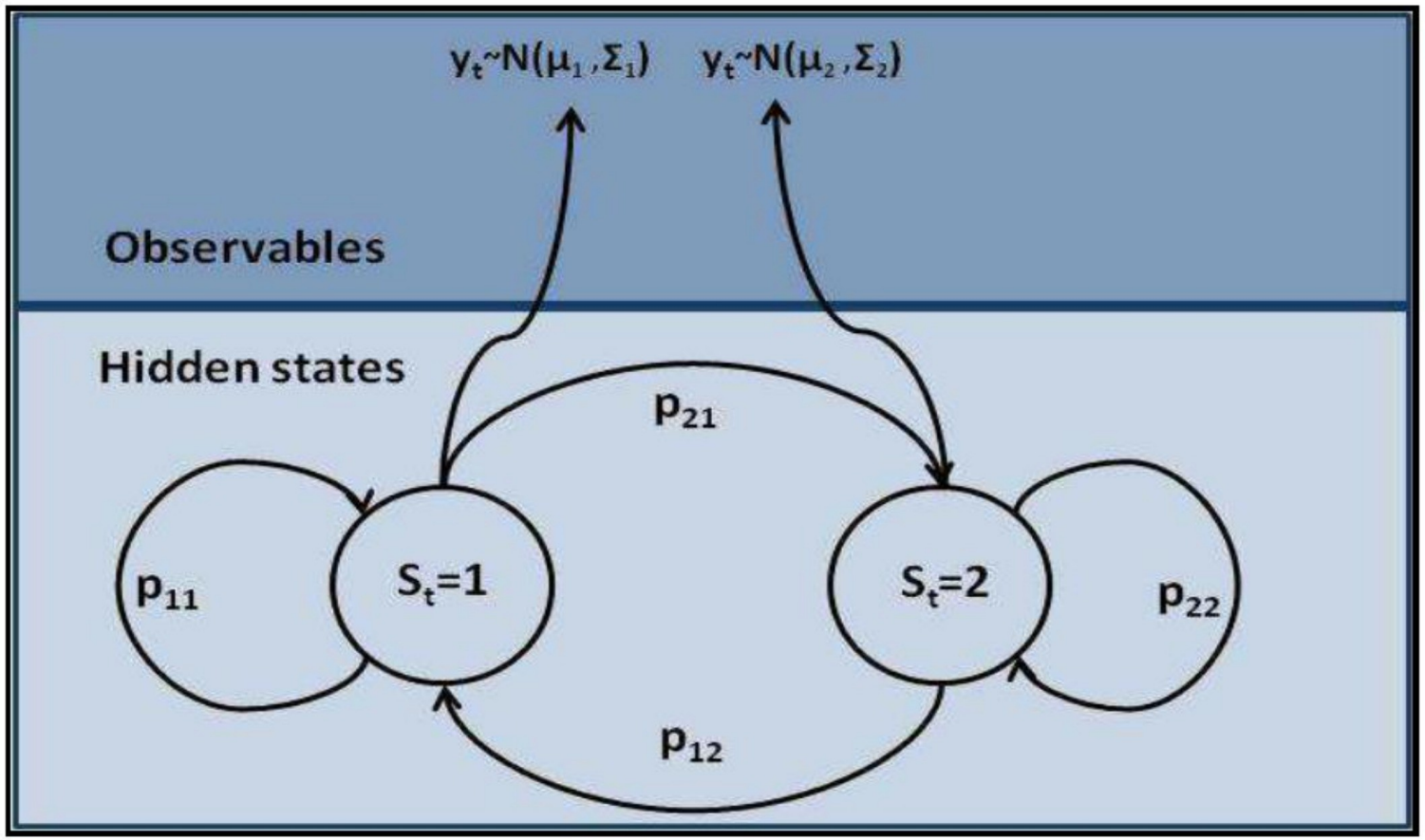


A two-state Gaussian HMM is the workhorse for return regimes

$y_t \sim N(\mu_k, \Sigma_k)$ in state k · p_{ij} = transition probabilities

Model

- S=1, S=2 hidden
- p11, p22: persistence
- Emissions: Normal
- Fit with Baum-Welch



02

PART 2 · STOCK PREDICTION

Stock Price Prediction with HMM

Model selection, price prediction and return regimes on AAPL and KOSPI

Information criteria choose the number of hidden states

Gaussian HMM, N states, M observations, parameters fitted by Baum-Welch

Read the page

- L: likelihood
- M: number of observations
- k: number of parameters
- Paper: states 2–4 only

Choosing the number of hidden states in a Hidden Markov Model (HMM) is an important task. In general, AIC (Akaike Information Criterion) and BIC (Bayesian Information Criterion) are used to evaluate model performance, which suits maximum likelihood estimation with the Baum-Welch algorithm. In this study, the number of states is limited to 2–4 to keep the model simple and practical.

AIC and BIC are computed as follows:

$$AIC = -2\ln(L) + 2k$$

$$BIC = -2\ln(L) + k\ln(M)$$

where:

- L is the likelihood function of the model,
- M is the number of observation points,
- k is the number of estimated parameters.

Each hidden state is assumed to be normally distributed. With N states in the HMM, the number of parameters k is defined as:

$$k = N^2 + 2N - 1$$

The simulation runs 100 times: each step uses the parameters estimated in the previous step as initial values, re-estimates, and computes AIC and BIC. This shows how BIC changes over the simulation.

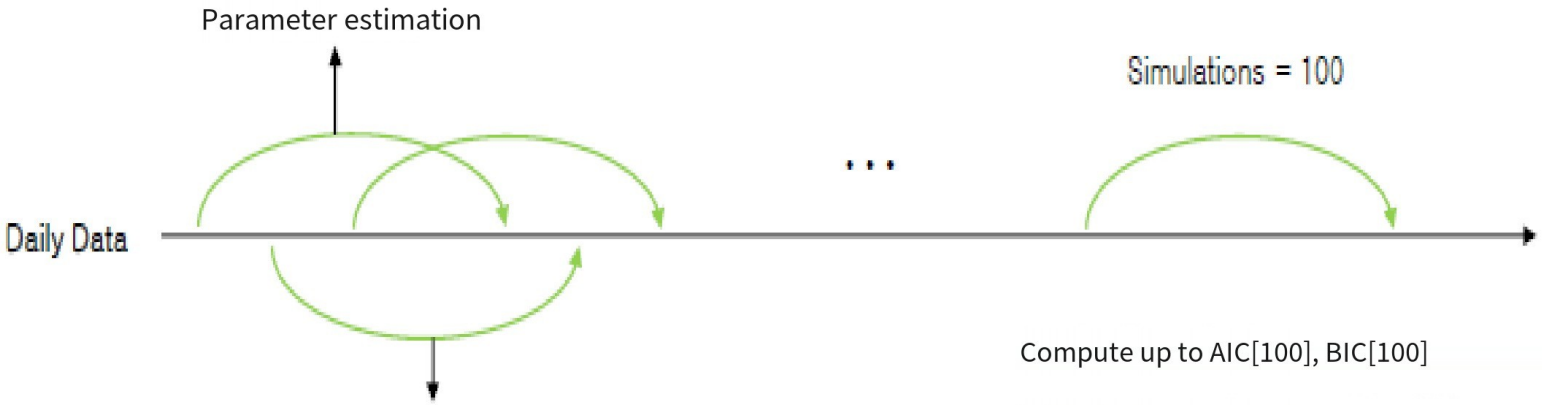
$$AIC = -2\ln(L) + 2k, \qquad BIC = -2\ln(L) + k\ln(M), \qquad k = N^2 + 2N - 1$$

Repeated calibration tracks AIC and BIC over 100 simulations

Each step starts from the previous step's parameter estimates

Calibration loop

- Slide one step forward
- Re-estimate from last λ
- Record AIC[i], BIC[i]
- Repeat 100 times



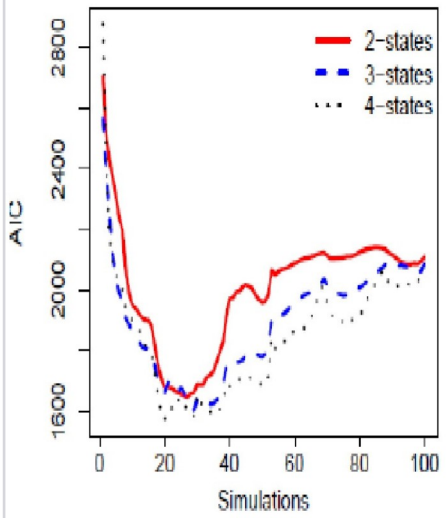
In the AAPL paper four states score lowest, and the gap persists on BIC

Apple daily data, 2015.1.6–2015.10.30 · left AIC, right BIC

Paper result

- AIC: 4 states lowest, gap narrows
- BIC: 4 states stay lowest
- Paper picks 4 states
- Not always U-shaped

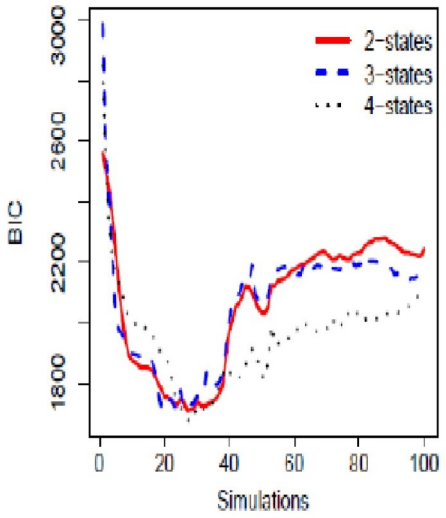
- Vary the number of states over 2, 3, 4 and measure AIC and BIC with the procedure on the previous page.
- The figure below is [Figure 1] of the paper: AIC and BIC of Apple for each number of states.
- For AIC, 4 states are smallest, but AIC converges across state counts as simulations increase, with little difference. For BIC, however, 4 states stay smallest as simulations increase, so 4 states can be regarded as suitable for Apple (per the paper).
- However, AIC and BIC do not always show this U-shape. The shape can differ by stock and analysis period.



Stock: Apple (AAPL) Daily data

Period: 2015.1.6 – 2015.10.30

Figure 1: AIC (left) and BIC (right) for 100 parameter calibrations of HMM using Apple, AAPL, stock daily trading prices



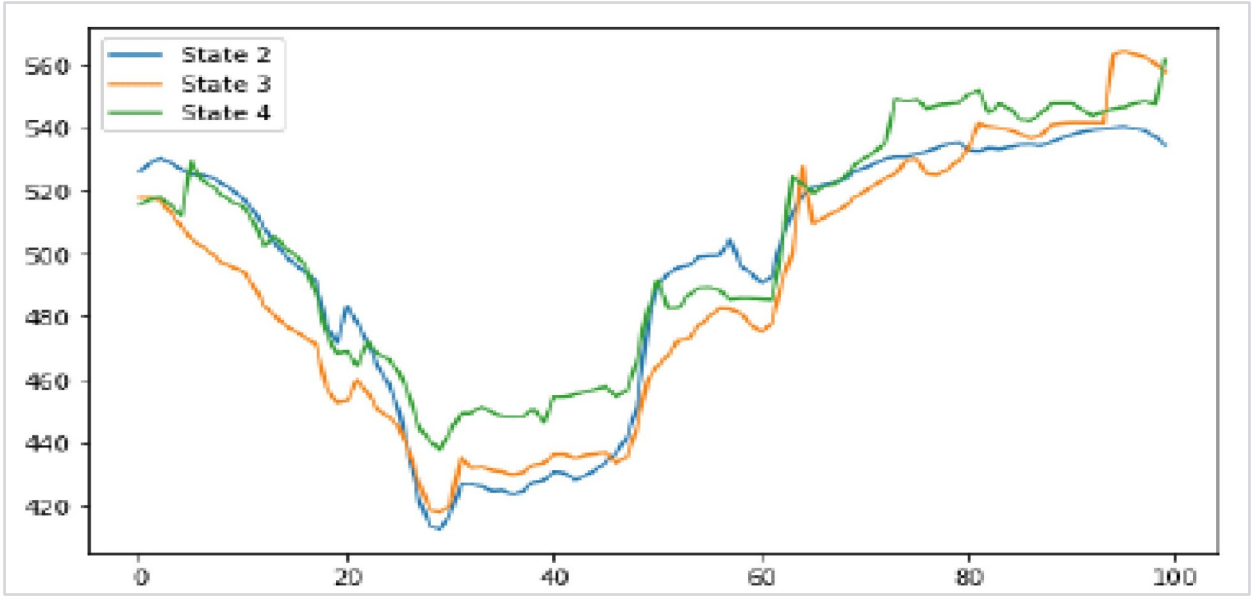
The BIC comparison is repeated on AAPL closing prices alone

hmmlearn, same period as the paper, 2–4 states

[Input] Prompt · Claude Code (Code tab)

Do the following and save the chart as bic_close.png.

1. Get AAPL daily closing prices, 2015-01-06 to 2015-10-30. Use the close only.
2. Fit a Gaussian HMM with hmmlearn for 2, 3 and 4 states.
3. Run 100 simulations: slide one step forward and re-estimate from the previous step's parameters.
4. At each step compute BIC = −2 ln(L) + k ln(M) with $k = N^2 + 2N - 1$.
5. Plot BIC per simulation, one curve per number of states.



[Check] What to confirm in the result

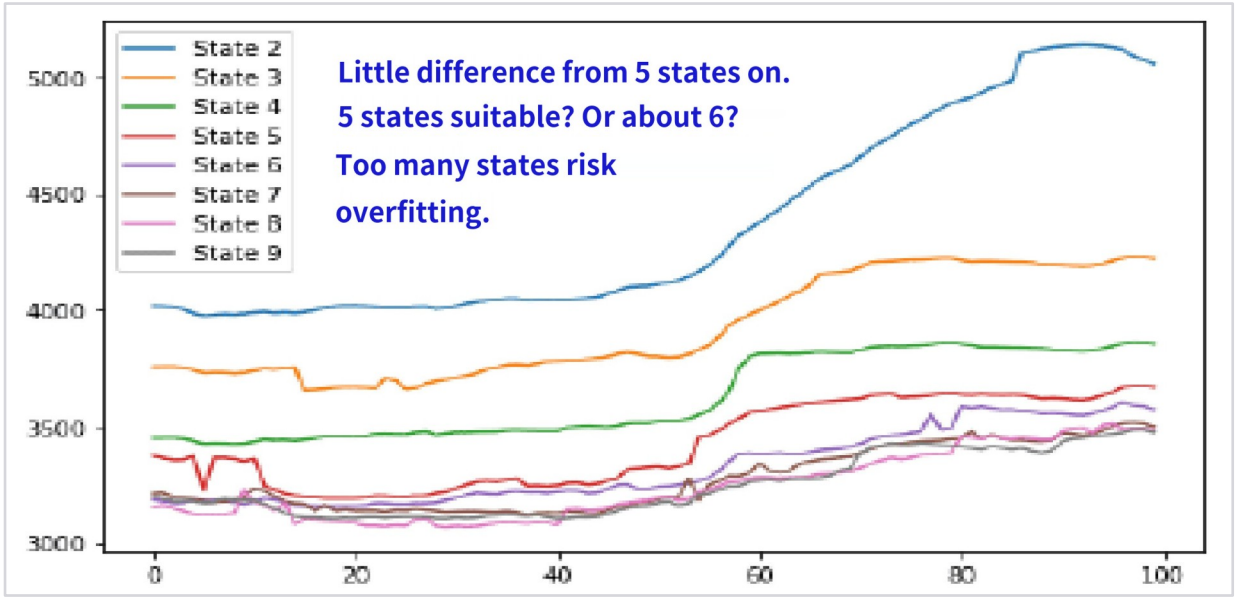
· Lower BIC = better
· Which number of states stays lowest?
· Compare with the paper's BIC on p.12

With OHLC inputs, gains in BIC fade beyond about five states

Open, high, low, close · 2–9 states · too many states risk overfitting

[Input] Prompt · Claude Code (Code tab)

```
Repeat the AAPL exercise and save the chart as
bic_ohlc.png.
1. Use the same period, with open, high, low
    and close as inputs.
2. Fit a Gaussian HMM for 2 to 9 states.
3. Run the same 100-simulation calibration loop
    and compute BIC at each step.
4. Plot BIC per simulation, one curve per
    number of states.
```



[Check] What to confirm in the result

· Little gain after 5–6
· Large N overfits

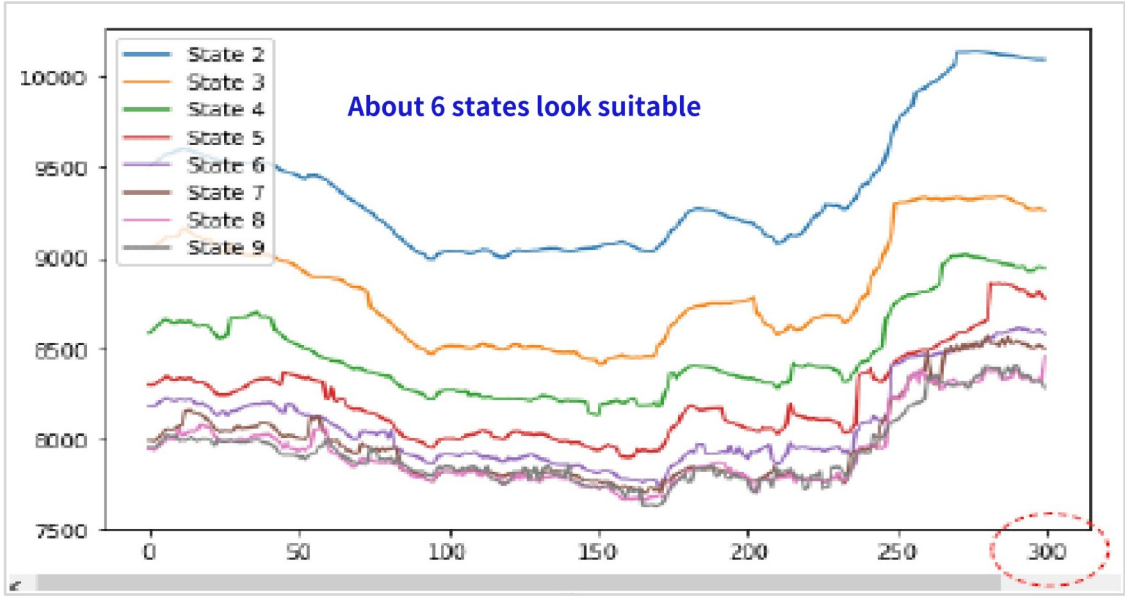
PRACTICE

KOSPI index data point to six or seven hidden states

Slide text says 7; the chart note says 6 — curves for 6–9 states nearly overlap

[Input] Prompt · Claude Code (Code tab)

```
Do the following and save the chart as
bic_kospi.png.
1. Get KOSPI composite index daily data. Tell
    me the inputs and period you used.
2. Fit a Gaussian HMM for 2 to 9 states.
3. Run 300 simulations with the same
    calibration loop and compute BIC at each
    step.
4. Plot BIC per simulation, one curve per
    number of states.
```



[Check] What to confirm in the result

· Curves converge at 6–7
· Pick the smaller N

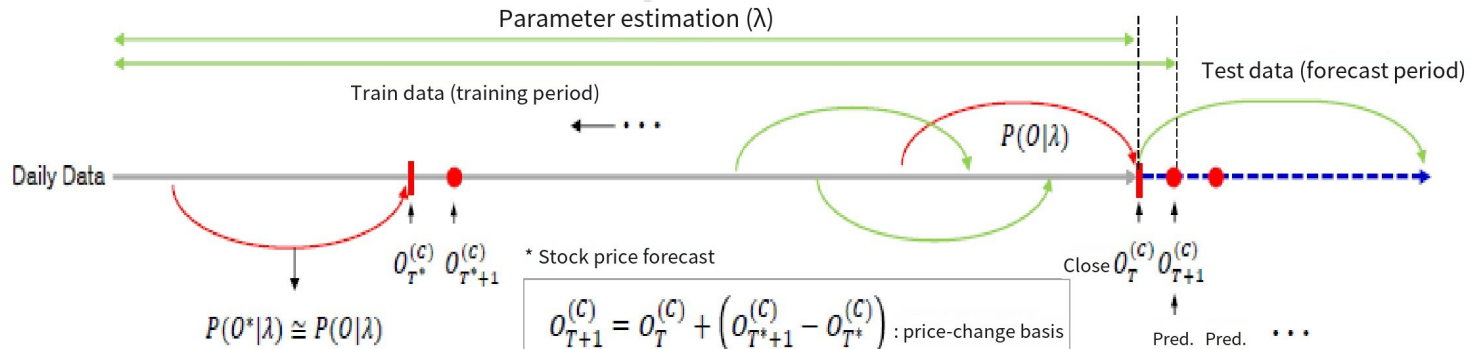
Prediction borrows the price change of the most similar past day

Train: fit λ · find day with likelihood closest to today · apply its change

Three steps

- 1 Calibrate λ, compute P(O|λ)
- 2 Find past day with similar P
- 3 Apply its change to today
- Two variants: price change or return

▪ We first introduce how to predict stock prices using HMM. The prediction process can be divided by three steps. Step one: calibrate HMM's parameters and calculate likelihood of the model. Step 2: find a day in the past that has a similar likelihood to that of the recent day. Finally, step 3: use the difference of stock prices on the "similar" day in the past to predict future stock prices. This prediction approach is based on the work of Hassan and Nath [2]. ~~However, our procedure is different from their method in that we put the sign of the difference of the likelihood of recent day with likelihood of the "similar" day in the past into the calculation of future stock prices~~. Furthermore, we do not only use HMM with multiple observations (open, low, high, close price), as in [2], but also the HMM with one observation data (close price) to predict future close price. By using both single and multiple observations to predict stocks' close prices, we can examine which method performed better. The HMM with a single observation is a special case of HMM with multiple observations, thus we will explain in detail how to use HMM with multiple observations.

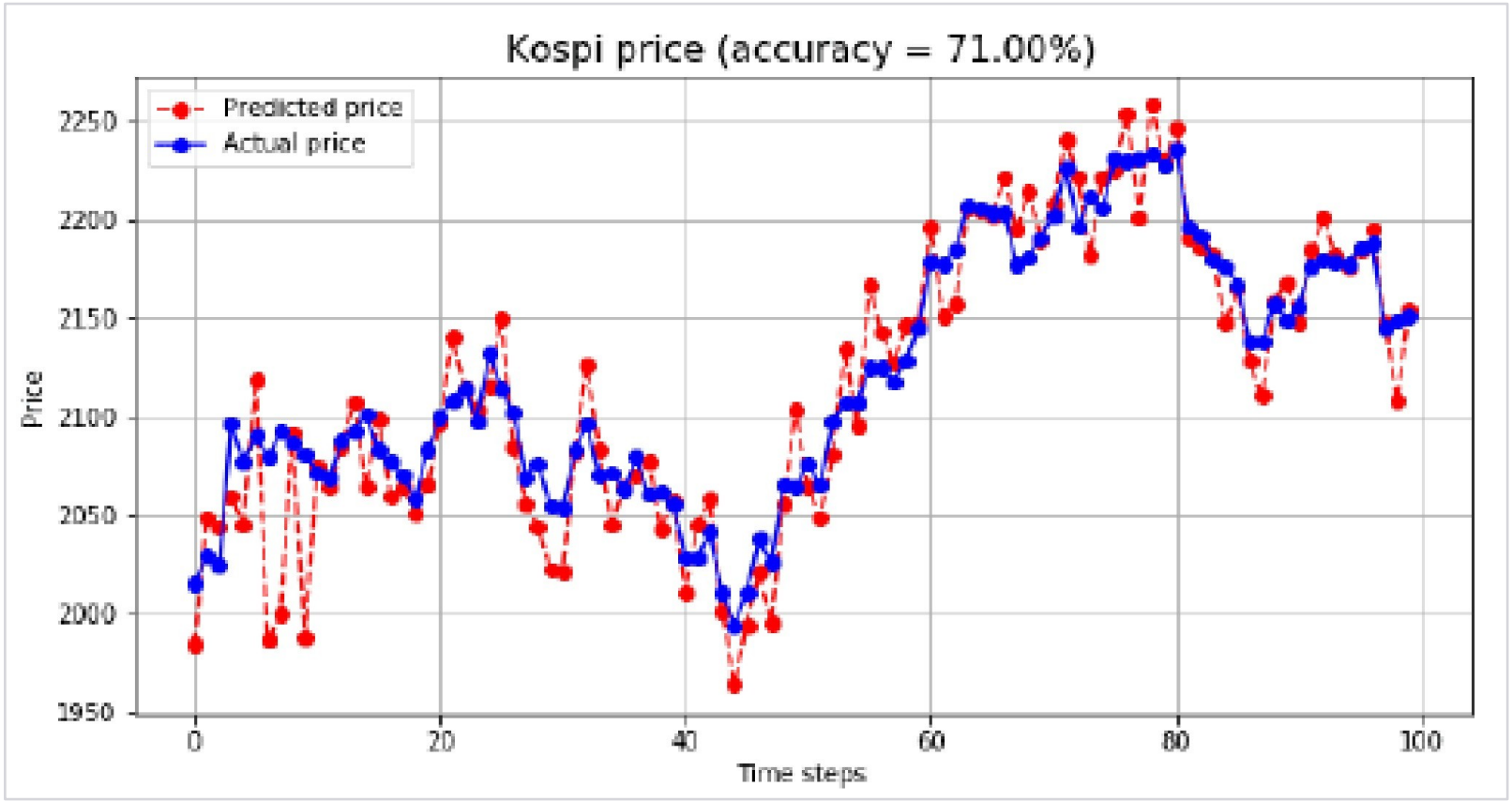


The rolling forecast tracks KOSPI over the last 100 days

Reported accuracy 71.00% on the test window

Result

- Red: predicted price
- Blue: actual price
- Accuracy = 71.00%
- Same logic → tomorrow



A three-state HMM splits KOSPI returns into distinct regimes

Estimated mean, volatility and transition matrix of each state

Estimates

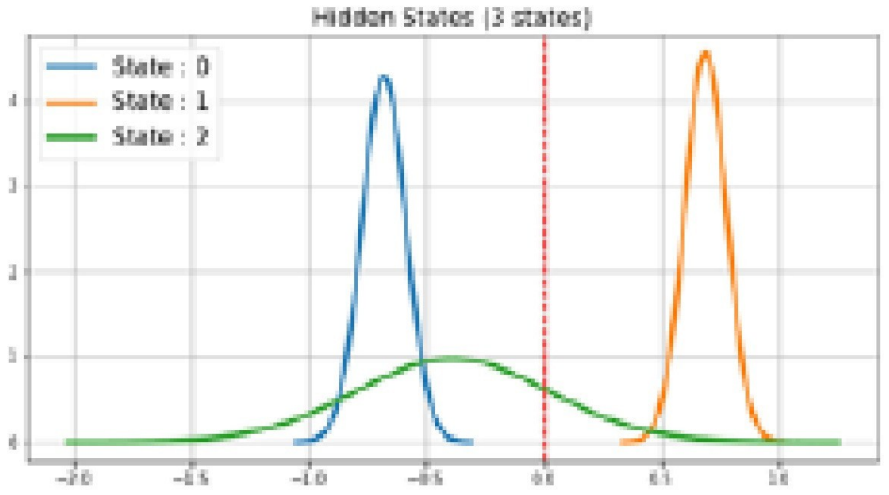
- μ: −0.681, 0.688, −0.389
- σ: 0.093, 0.088, 0.411
- Diagonal: 0.855, 0.90, 0.932
- Regimes are persistent

```
mu :
 [[-0.681]
 [ 0.688]
 [-0.389]]

vol :
 [[0.093]
 [0.088]
 [0.411]]

Transition :
 [[0.855 0.118 0.027]
 [0.099 0.9   0.002]
 [0.017 0.051 0.932]]
```

Mean and std. dev. of the 3 return distributions



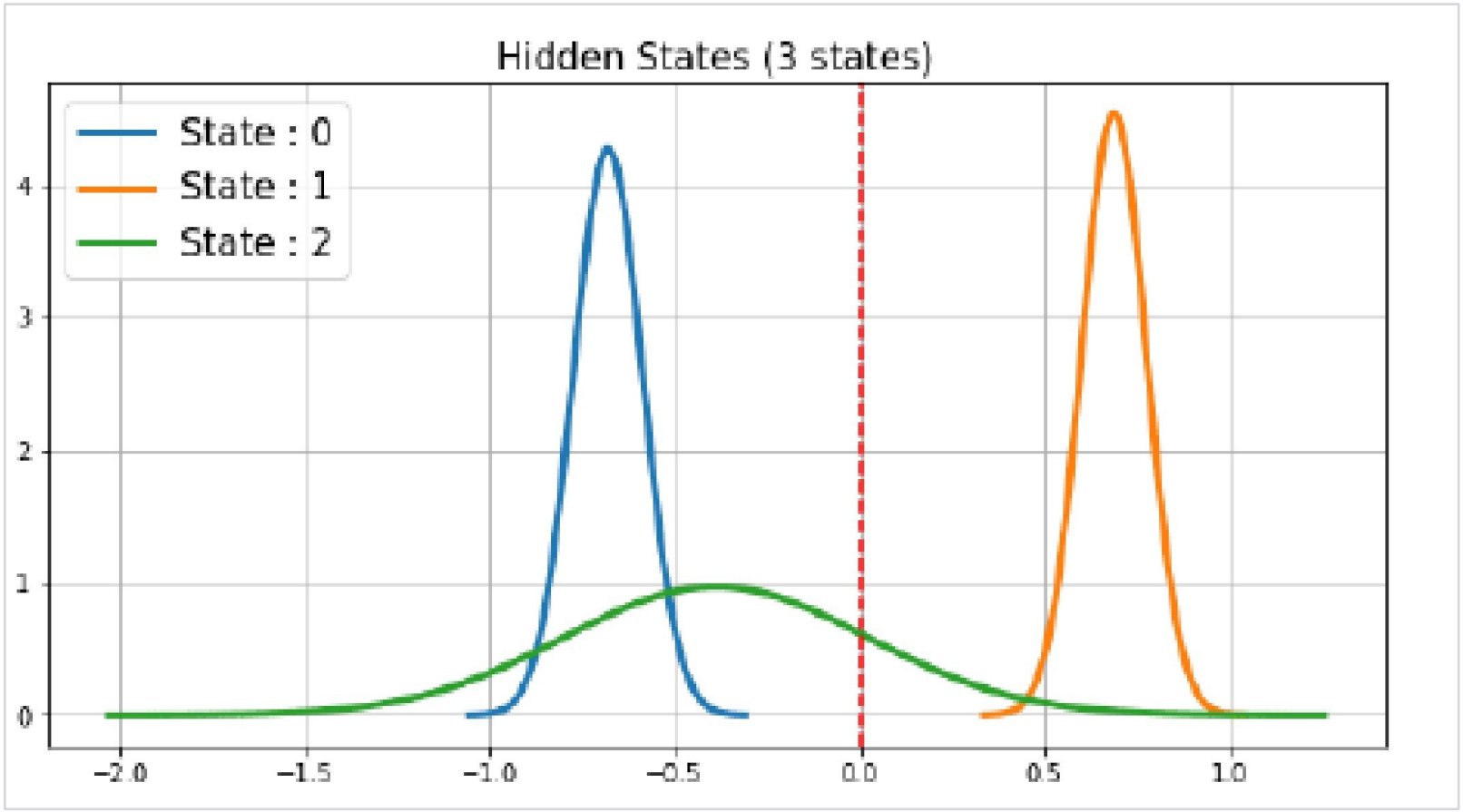
Transition probabilities of the 3 return distributions

The return distributions show two calm states and one volatile state

State 0: negative mean · State 1: positive mean · State 2: wide spread

Read the curves

- Blue: calm down market
- Orange: calm up market
- Green: high volatility
- Red line: zero return



The KOSPI closing price of the last 300 days is the reference picture

2018-01 to 2019-03 daily close

Look first

- Peak in early 2018
- Fall into late 2018
- Partial rebound in 2019

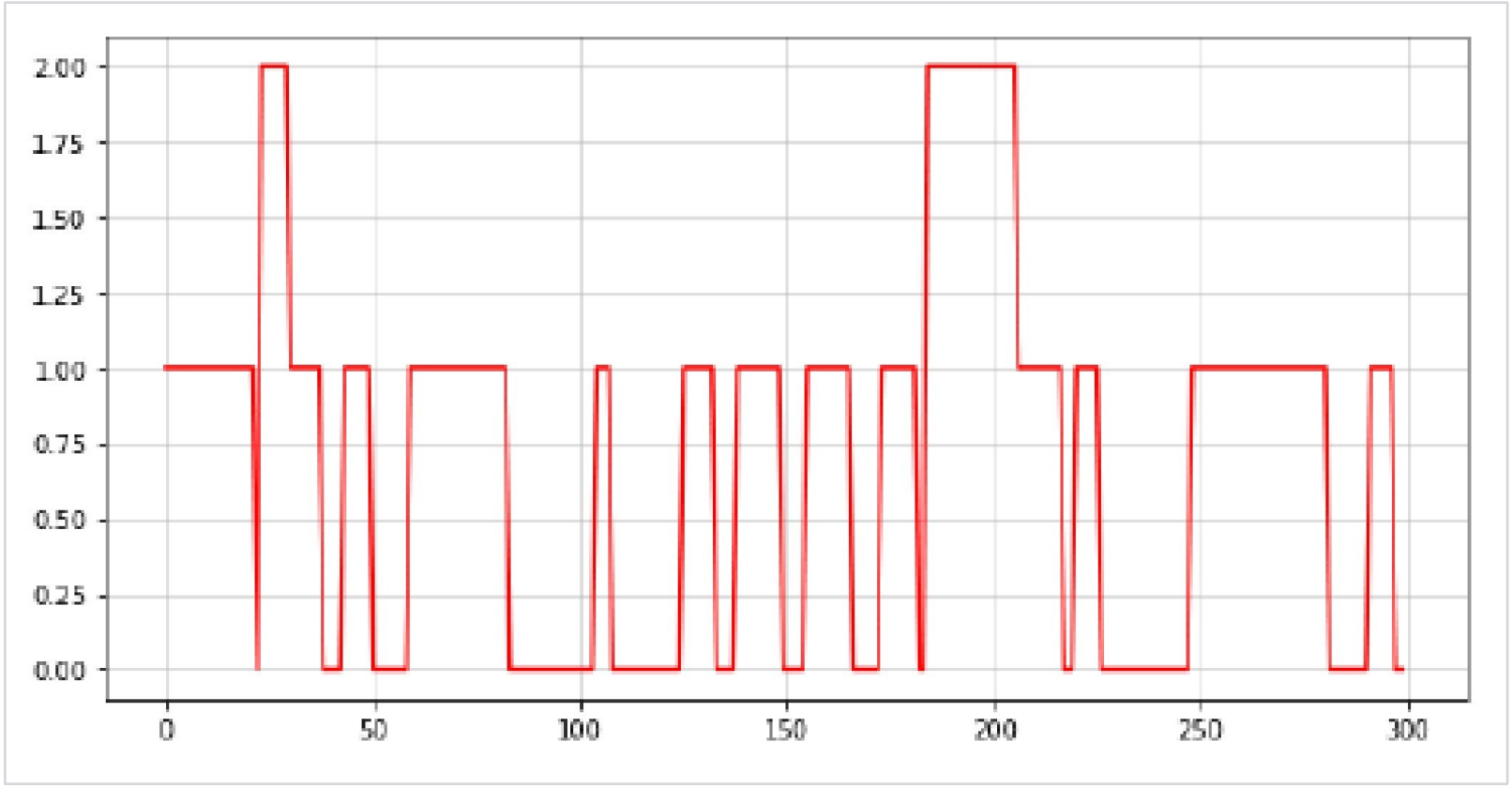


The decoded state path marks when the market switched regime

Last 300 days · current state 0 (mean −0.681, std 0.093 as stated in the slide)

Read the path

- y-axis: state 0, 1, 2
- Spikes to 2: stress
- Now: state 0



Parameters, distributions and the state path tell one story

Summary of the KOSPI return-regime exercise

Put together

- Table: μ, σ, A
- Curves: per-state returns
- Top right: state path
- Bottom right: price

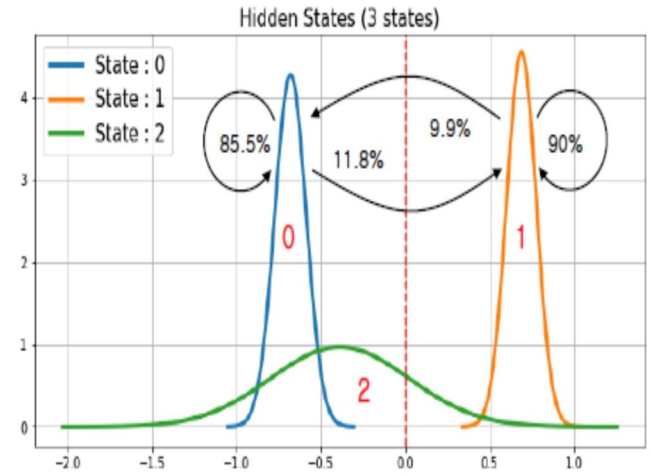
☐ Stock return state-change exercise

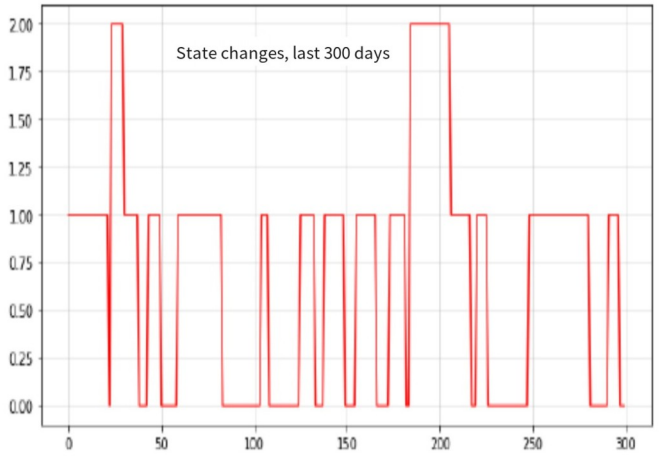
▪ States

State	Mean	Std. dev.
0	−0.681	0.093
1	0.688	0.088
2	−0.389	0.411

▪ Transition probability

State	0	1	2
0	0.855	0.118	0.027
1	0.099	0.9	0.002
2	0.017	0.051	0.932







03

PART 3 · RATES CARRY

Rates Carry Strategy with HMM

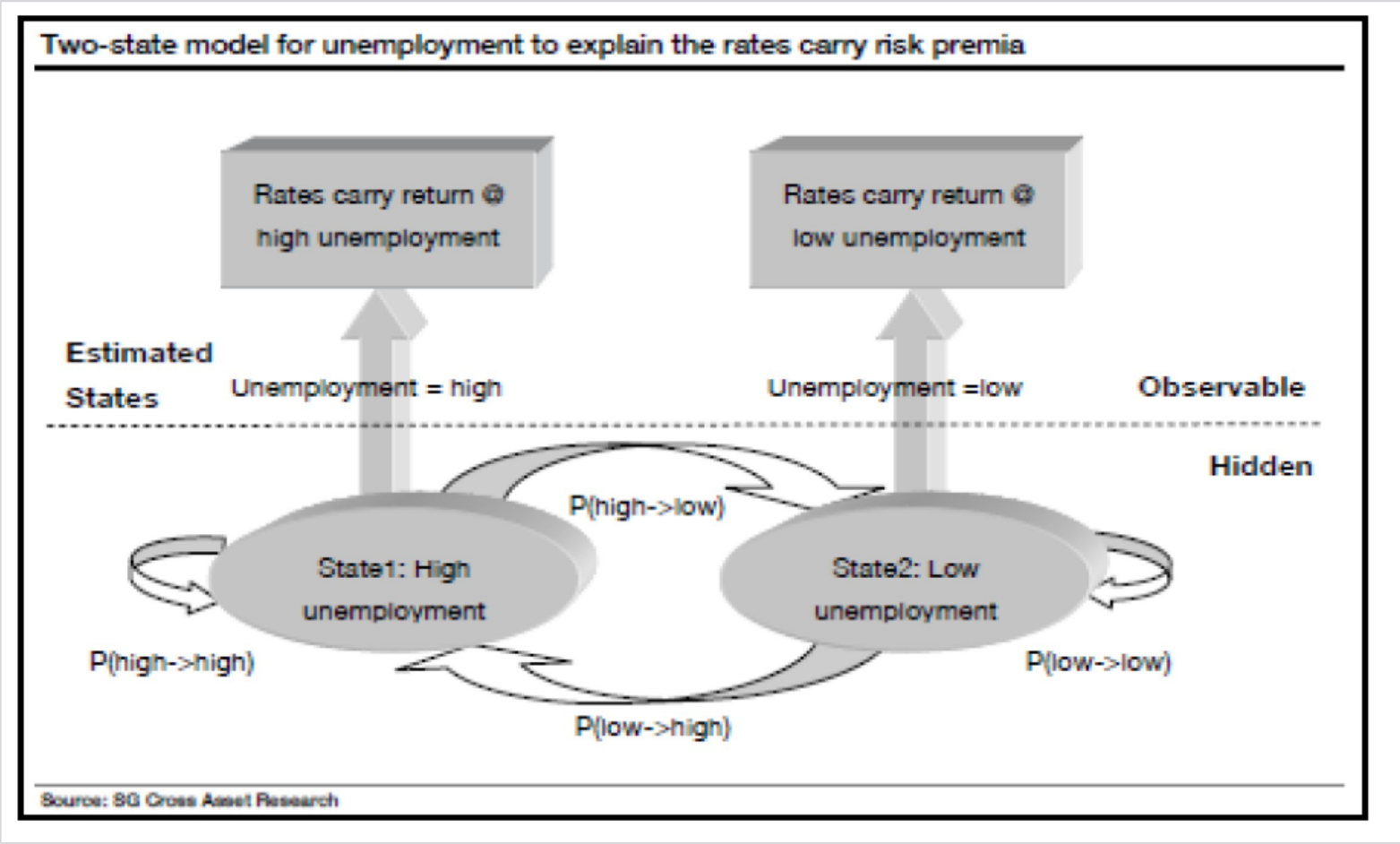
Unemployment regimes and the carry risk premium

Unemployment regimes explain when rates carry pays

Two-state model: high vs low unemployment (SG Cross Asset Research)

Idea

- Hidden: unemployment regime
- Observed: carry return
- Return differs by regime
- Use state to time carry



04

PART 4 · INFORMED TRADING

An HMM Approach to Information-Based Trading

Kyle model, volume time, PIN and VPIN

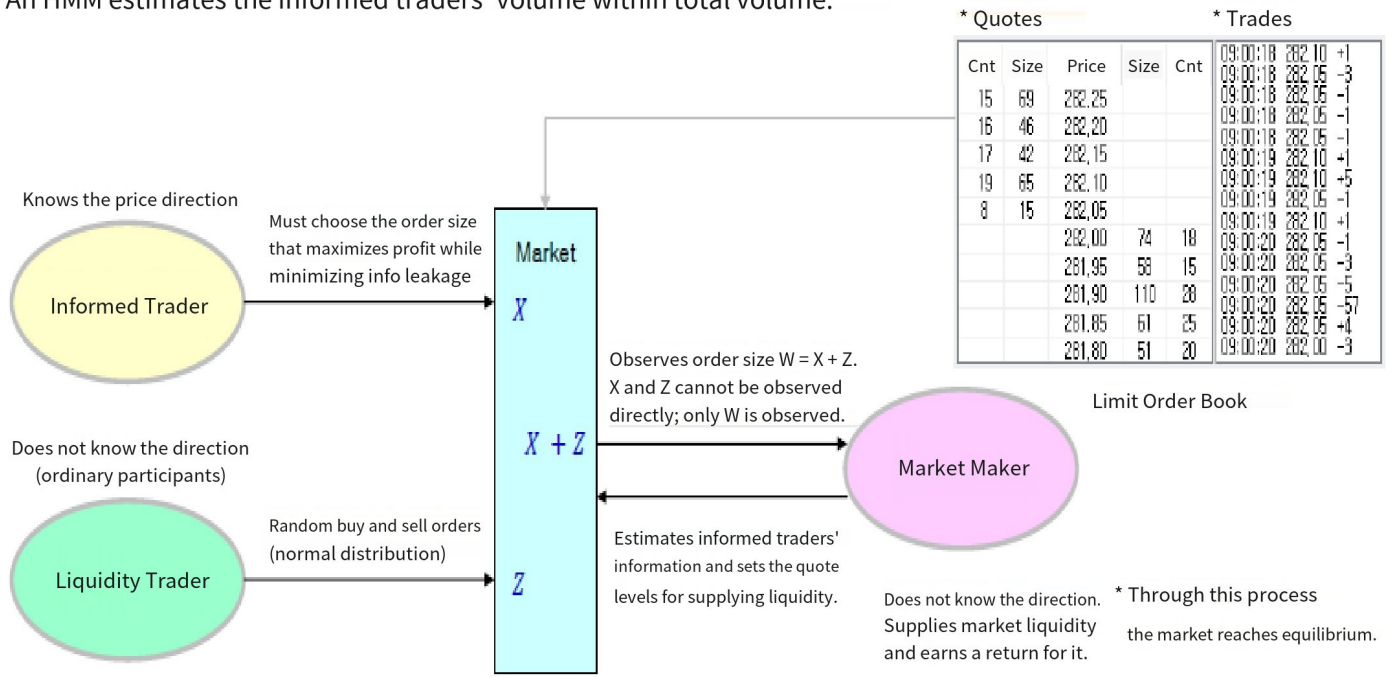
Kyle's model has three players: informed, noise, market maker

Informed trader · liquidity trader · market maker

Players

- Informed: private info
- Liquidity: noise orders
- Market maker: sets price
- HMM: share of informed volume

- Model of a market with Informed Traders, Liquidity Traders and a Market Maker.
- Informed traders hold private information and know whether the price will rise or fall (assumption).
- Liquidity traders have no information and trade for their own purposes (hedging) (Noise Traders). Their buy and sell orders are random.
- The market maker posts limit buy and sell orders on the Ask and Bid sides at once. These resting orders supply liquidity to the market.
- An HMM estimates the informed traders' volume within total volume.



Tick-time bars adapt to activity where clock-time bars cannot

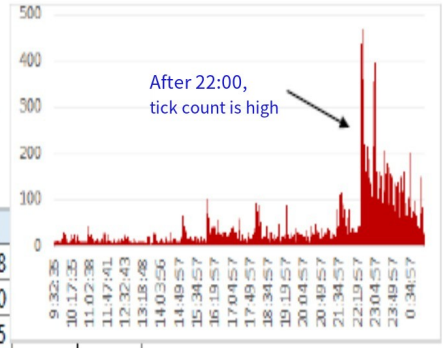
E-mini S&P 500 futures · clock time vs tick time candles

Clock vs tick

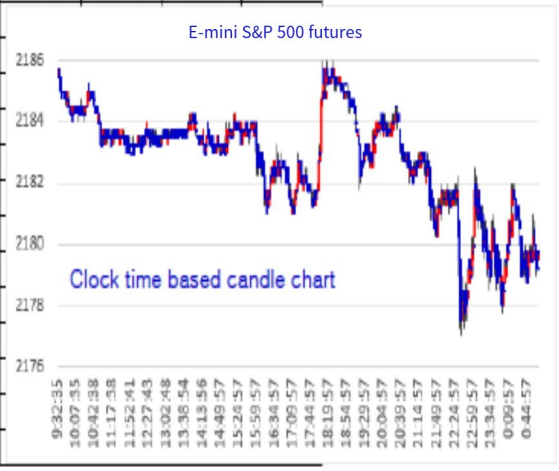
- Clock: fixed time
- Tick: fixed trade count
- Activity spikes after 22:00
- Tick bars stay informative

- A standard candle chart records prices (open, high, low, close) over a fixed time. → Clock time based candle chart.
- Market information enters through volume, so all information is reflected in the price. → The price is a vehicle of information.
- But volume varies by time of day, so each Clock-time candle carries a different amount of information.

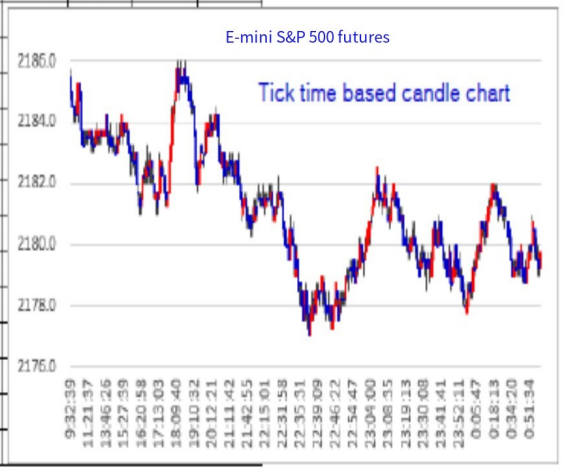
  For example, volume is high right after the open and before the close and low at lunchtime, so candles formed right after the open cannot be treated the same as candles formed at lunchtime.
- Forming a candle every fixed number of executed ticks gives a Tick time based candle chart.



time	ticks	open	high	low	close
9:32:35	8	2185.5	2185.75	2185.5	2185.5
9:33:35	5	2185.5	2185.75	2185.5	2185.75
9:34:35	6	2185.5	2185.5	2185.5	2185.5
9:35:35	11				
9:36:35	5				
9:37:35	8				
9:38:35	14				
9:39:35	14				
9:40:35	9				
9:41:35	12				
9:42:35	3				
9:43:35	11				
9:44:35	1				
9:45:35	3				
9:46:35	8				
9:47:35	6				



time	ticks	open	high
9:32:39	100	2185.5	2185.8
9:43:50	100	2185.0	2185.0
9:55:48	100	2184.5	2184.5
10:01:41	100		
10:14:49	100		
10:23:12	100		
10:31:39	100		
10:42:02	100		
10:54:45	100		
10:57:41	100		
11:03:49	100		
11:12:24	100		
11:21:37	100		
11:32:22	100		
11:41:19	100		
11:50:13	100		



Volume-time bars close after a fixed traded volume

EUR/USD futures, 2016-08-16 · 1,500 lots per bar · buy/sell ratio per bar

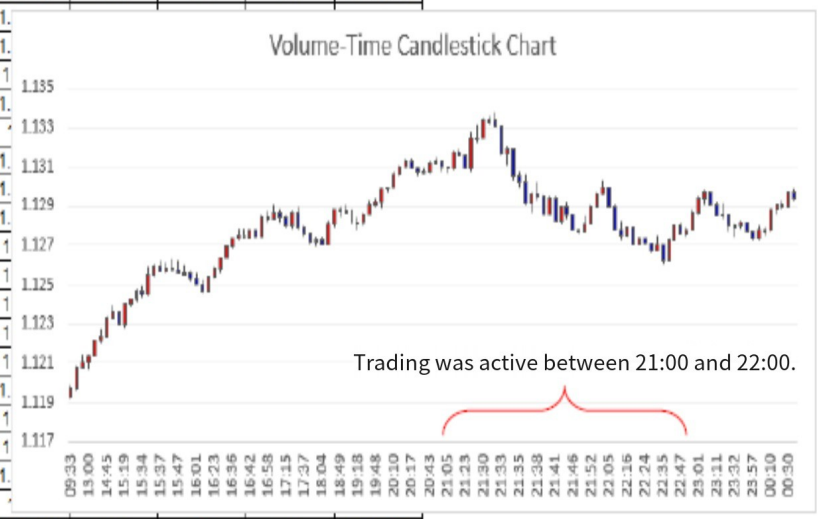
Volume time

- Bar = fixed volume
- Record buy and sell
- Ratio B/S per bar
- Input for the HMM

- A candle formed every fixed traded volume is called a Volume time based candle chart.

Euro Currency Futures EUR/USD (2016.8.16 09:32:39 ~ next day 01:08:00)

date	time	ticks	open	high	low	close	volume	buy	sell	buy/sell
2016-08-16	09:33	475	1.11925	1.1199	1.1192	1.1197	1500	795	705	1.1276596
2016-08-16	11:38	376	1.1197	1.12085	1.1196	1.1208	1500	962	538	1.7881041
2016-08-16	12:20	332	1.1208	1.12145	1.12075	1.121	1500	837	663	1.2624434
2016-08-16	13:00	280	1.121	1.1214	1.1207	1.12135	1500	956	544	1.7573529
2016-08-16	13:58	290	1.12135	1.1222	1.12135	1.				
2016-08-16	14:17	396	1.12215	1.12275	1.122	1.				
2016-08-16	14:45	388	1.12235	1.12335	1.1223	1				
2016-08-16	15:05	238	1.1233	1.12395	1.1233	1.				
2016-08-16	15:07	295	1.12365	1.12365	1.123					
2016-08-16	15:19	246	1.123	1.12405	1.12285	1.				
2016-08-16	15:31	164	1.12405	1.1243	1.1239	1.				
2016-08-16	15:33	186	1.12425	1.1248	1.1241	1.				
2016-08-16	15:34	219	1.12465	1.1249	1.12435	1				
2016-08-16	15:36	159	1.1245	1.1259	1.12445	1				
2016-08-16	15:36	280	1.1255	1.12605	1.1254	1				
2016-08-16	15:37	290	1.1259	1.12625	1.12565	1				
2016-08-16	15:39	327	1.1257	1.12615	1.1256	1				
2016-08-16	15:42	288	1.1258	1.12635	1.12565	1.				
2016-08-16	15:47	259	1.12575	1.12625	1.1256	1				
2016-08-16	15:53	258	1.1256	1.126	1.1255	1				
2016-08-16	15:58	231	1.1256	1.1257	1.12515	1.				
2016-08-16	16:01	318	1.12535	1.1256	1.125					
2016-08-16	16:07	305	1.125	1.12525	1.1246	1.12465	1500	600	900	0.6666667
2016-08-16	16:14	344	1.12465	1.1254	1.12465	1.1254	1500	891	609	1.4630542
2016-08-16	16:23	215	1.1254	1.1259	1.1253	1.12585	1500	1075	425	2.5294118
2016-08-16	16:28	348	1.12585	1.1264	1.1256	1.1264	1500	955	545	1.7522936



An HMM infers the informed-trading share from buy and sell flow

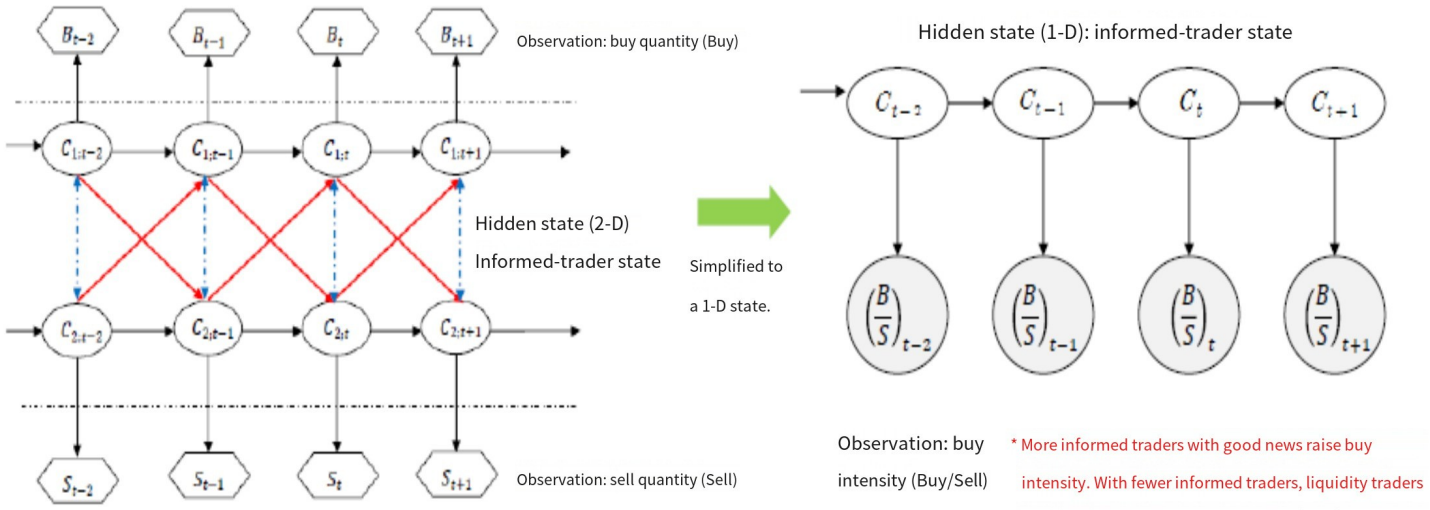
Yin & Zhao (2012), A Hidden Markov Process Approach to Information-Based Trading

Model

- Hidden: informed share
- Observed: buys, sells
- 2-D → 1-D state
- Observation: B/S

HMM inference of the hidden state (share) of informed traders

- Estimate the informed traders' share by observing buy and sell order quantities in the market, or buy order intensity (Buy / Sell).
- The informed traders' participation share can be viewed as a hidden state of order quantity. If they hold good news, buy orders will exceed sell orders; if they hold bad news, sell orders will dominate. With no informed traders (no information entering the market), only Noise Traders remain, so buy and sell orders arrive randomly.
- In market microstructure, models of informed participation include PIN (Probability of Informed Trading) and VPIN (Volume Synchronized PIN); PIN can also be estimated with an HMM.



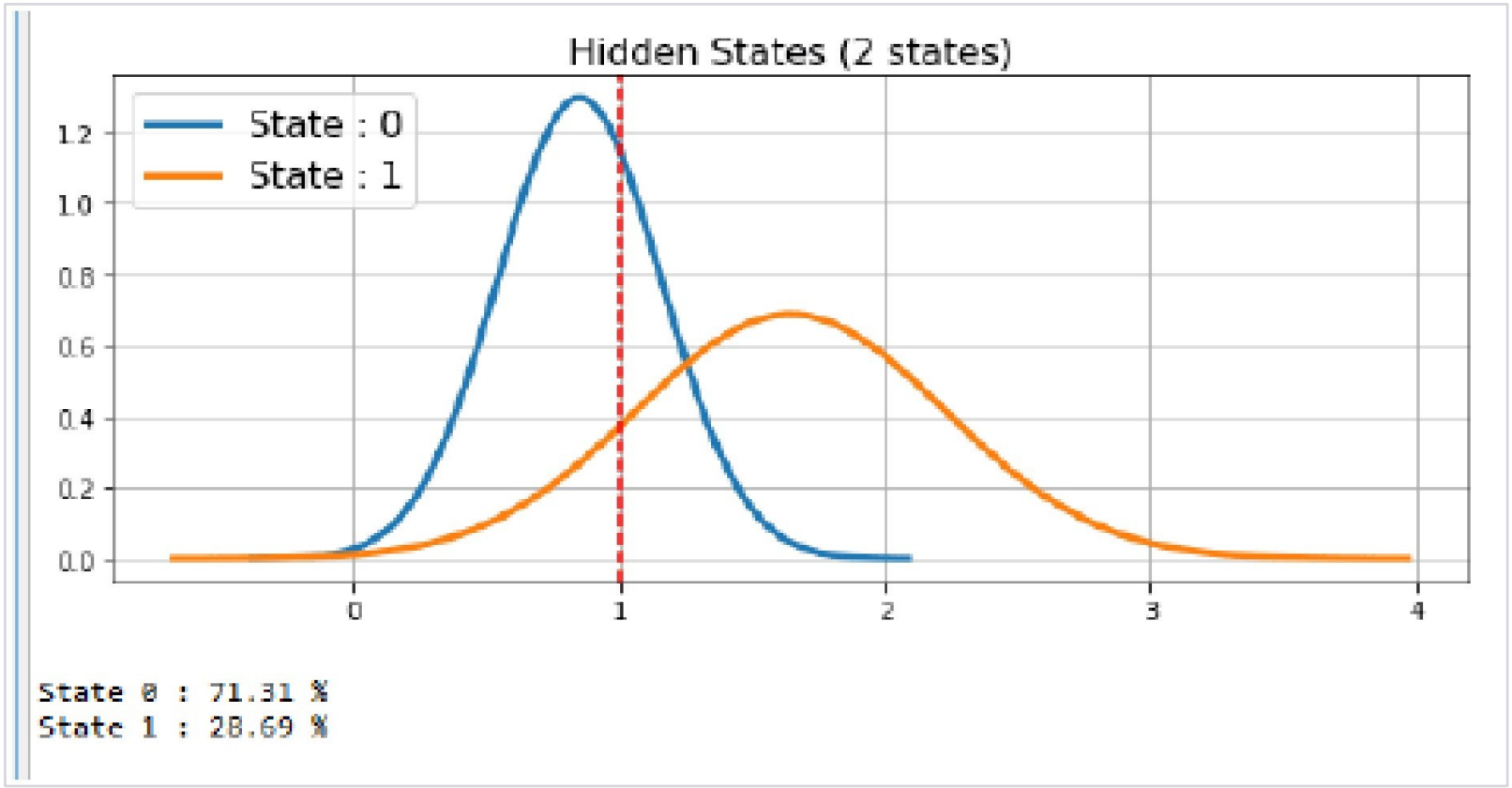
* Figure source: Xiangkang Yin, Jing Zhao, 2012, A Hidden Markov Process Approach to Information-Based Trading. (Figure 2)

Two states separate ordinary flow from buy-heavy flow

Gaussian HMM on B/S · state 1 (mean B/S > 1) covers 28.69% of bars

Two states

- State 0: 71.31%
- State 1: 28.69%
- Red line: B/S = 1
- State 1 = informed buying



Informed orders appear as short bursts in the state path

Transition: stay 0.841 · switch to informed 0.159 (slide text: 0.84 / 15.9%)

Read the path

- Now: state 0
- Stay prob. 0.841
- Switch prob. 0.159
- Circles: informed bursts

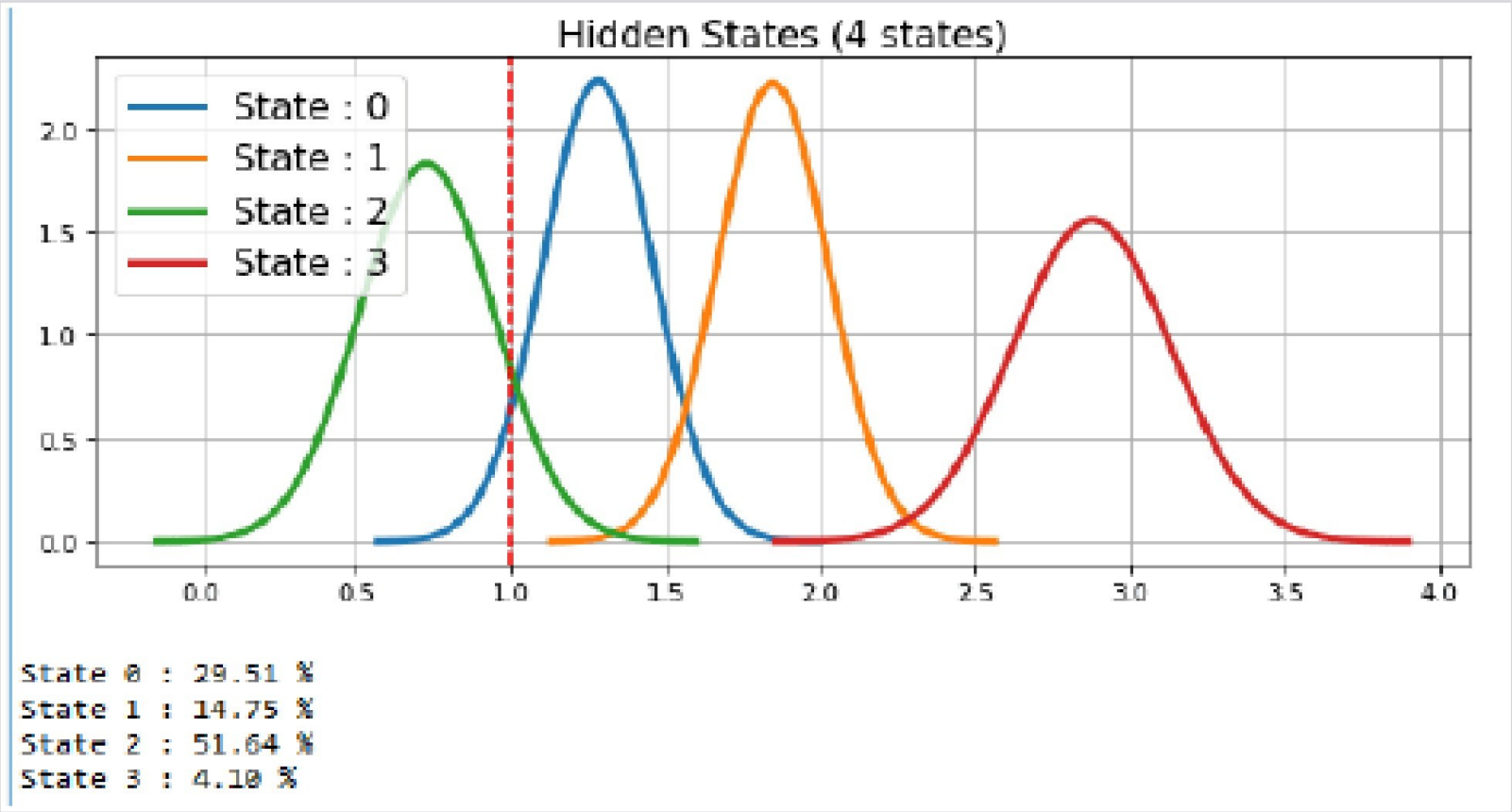


Four states describe order flow without a structural model

No Kyle or PIN assumptions are needed for the HMM

Four states

- State 0: 29.51%
- State 1: 14.75%
- State 2: 51.64%
- State 3: 4.10%

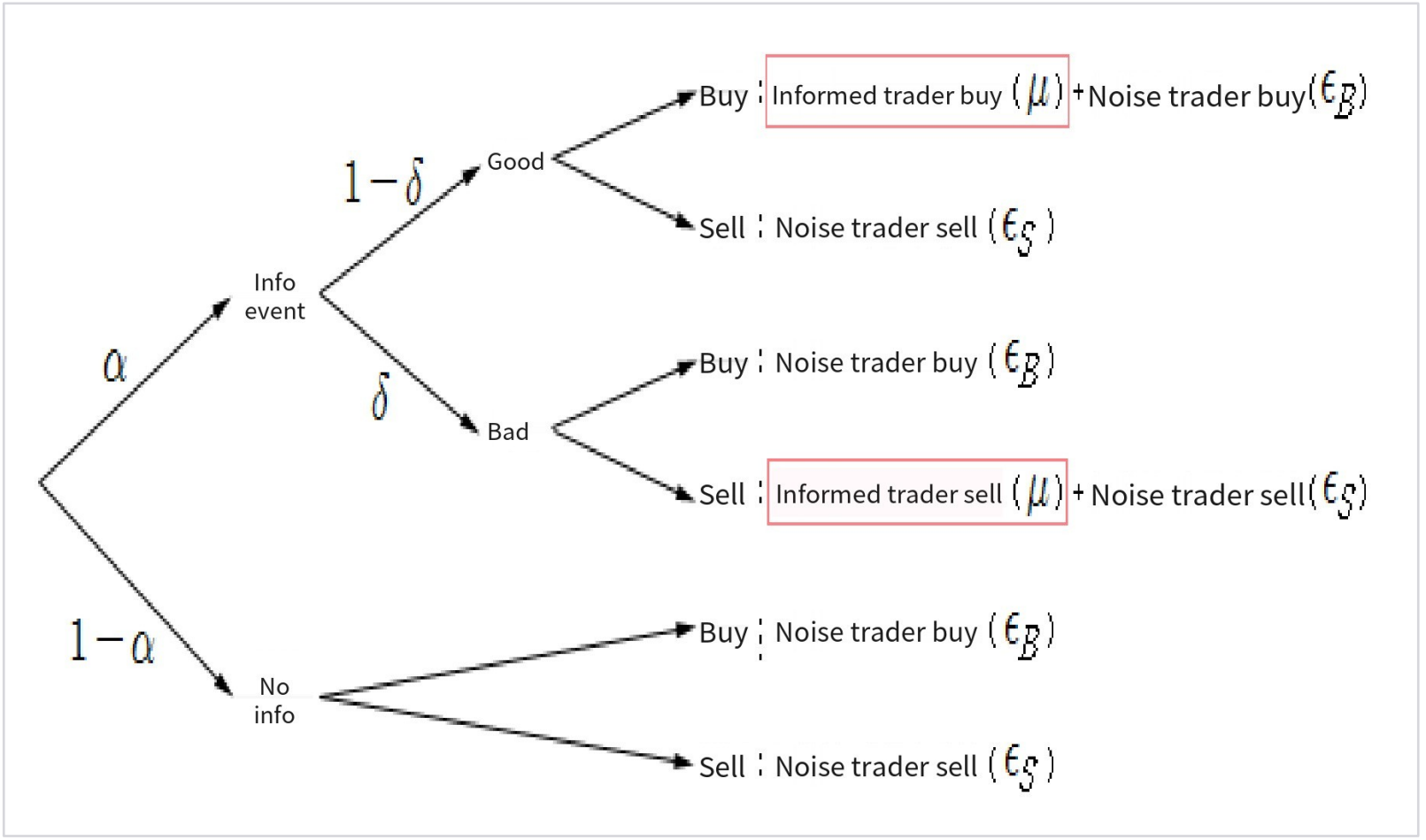


PIN models trade arrivals as a tree of information events

Easley et al. (1996) · information event · good news · bad news

Event tree

- α: info event occurs
- δ: news is bad
- μ: informed arrival
- ε_B, ε_S: noise buys, sells



PIN is the informed share of total trading intensity

Derivation on the original slide: buy, sell and total intensities

Steps

- Buy: α(1−δ)μ + ε_B
- Sell: αδμ + ε_S
- Total: αμ + ε_B + ε_S
- PIN = αμ / total

1) Buy intensity $= \alpha(1-\delta)(\mu+\epsilon_B) + \alpha\delta\epsilon_B + (1-\alpha)\epsilon_B$

$= \alpha(1-\delta)\mu + \epsilon_B$ ← Total buy intensity observed in the market

2) Sell intensity $= \alpha(1-\delta)\epsilon_S + \alpha\delta(\mu+\epsilon_S) + (1-\alpha)\epsilon_S$

$= \alpha\delta\mu + \epsilon_S$ ← Total sell intensity observed in the market

3) Total intensity $= \alpha(1-\delta)\mu + \epsilon_B + \alpha\delta\mu + \epsilon_S$

$= \alpha\mu + \epsilon_B + \epsilon_S$ ← Total buy intensity + total sell intensity

4) Total intensity of informed traders $= \alpha\mu$ ← Informed traders trade only when they have information

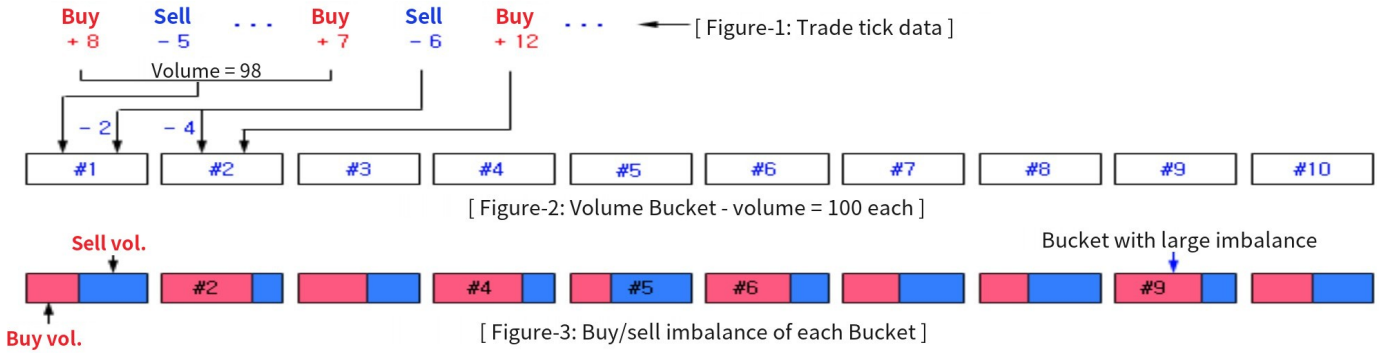
5) $PIN = \dfrac{\text{Total intensity of informed traders}}{\text{Total intensity}} = \dfrac{\alpha\mu}{\alpha\mu + \epsilon_B + \epsilon_S}$

VPIN estimates PIN from buy-sell imbalance in volume buckets

Each bucket holds the same volume V; imbalance |S − B| proxies αμ

Steps

- Group trades in buckets
- Measure |S_i − B_i|
- Average and divide by V
- VPIN ≈ PIN



1) $PIN = \frac{\alpha\mu}{\alpha\mu + 2\epsilon}$

2) $\alpha\mu \approx E[|S - B|] = \frac{1}{n}\sum_{i=1}^{n}|S_i - B_i|$ ← n = total number of Volume Buckets; S_i = Sell quantity of the i-th Volume Bucket; B_i = Buy quantity of the i-th Volume Bucket

3) $\alpha\mu + 2\epsilon = E[S + B] = V$ ← Constant, because each Bucket's volume (V) is kept fixed

4) $PIN = \frac{\alpha\mu}{\alpha\mu + 2\epsilon} \approx \frac{1}{nV}\sum_{i=1}^{n}|S_i - B_i| = VPIN$

5) $VPIN = \frac{1}{nV}\sum_{i=1}^{n}|S_i - B_i|$

PIN and VPIN reduce to a ratio of informed to total flow

Formulas from the original slides

$$PIN = \frac{\alpha\mu}{\alpha\mu+\epsilon_B+\epsilon_S} \qquad VPIN = \frac{1}{nV}\sum_{i=1}^{n}|S_i - B_i|$$

PIN

- Needs α, δ, μ, ε by MLE
- Daily buy/sell counts

VPIN

- Volume buckets, no MLE
- Works at high frequency

Order-flow imbalance counts bid and ask changes at the top of the book

Cont et al. (2010) · event-level e_n aggregated over interval k

OFI

- Bid-side event: + q_B
- Ask-side event: – q_A
- Sum e_n over interval
- Another HMM input

Alternative, event-based (Level-1) OFI as per (Cont et al., 2010) is based on order book events:

$$e_n = I_{\{P_n^B \geq P_{n-1}^B\}} q_n^B - I_{\{P_n^B \leq P_{n-1}^B\}} q_{n-1}^B - I_{\{P_n^A \leq P_{n-1}^A\}} q_n^A + I_{\{P_n^A \geq P_{n-1}^A\}} q_{n-1}^A$$

Aggregate over interval k:

$$\mathrm{OFI}_k = \sum_{n=N(t_{k-1})+1}^{N(t_k)} e_n$$

A few statements in the original slides need care

Figures follow the original slides; notes below apply when reading them

Page	As printed in the original	Note
p.15	Text: about 7 states	Chart note says 6; both are defensible
p.21	Annual mean −0.681, std 0.093	Same numbers as state-0 μ, σ on p.18; unit not stated
p.28	Table date 2016-08-01	Corrected to 2016-08-16, as in the table title
p.31	0.84 and 15.9%	Matrix shows 0.841 and 0.159

DISCUSSION

Four questions connect the HMM output to investment decisions

Use the KOSPI and EUR/USD examples

Q1	AIC prefers more states than BIC. Which would you use for a live trading model, and why?	**Model choice**
Q2	The forecast reports 71% accuracy over 100 days. What else would you check before trading it?	**Prediction**
Q3	State 2 on KOSPI has $\sigma = 0.411$. How would you change exposure when the model enters it?	**Regimes**
Q4	Why can an HMM on B/S find informed trading without the PIN tree assumptions?	**Microstructure**

Hidden states turn market data into regime signals

Choose N, decode the path, act on the state

1

Choose the number of states with AIC and BIC

Gains fade after about 5–7 states on AAPL OHLC and KOSPI; more states overfit.

2

Read regimes from the decoded path

KOSPI splits into calm-down, calm-up and high-volatility states with persistent transitions.

3

Detect informed flow from buy-sell imbalance

A two-state HMM on B/S flags 28.69% of volume bars as buy-heavy; PIN and VPIN give the structural view.

Next: the HMM algorithms step by step with applications (HMM Applications deck).